



LAB REPORT 01

JAVA

Submitted by,

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Course : Introduction to Programming
Language II (Java) Lab
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A line drawing of a pen writing the signature 'Bayazid' in a cursive script.

Bayazid

Problem 01

-Creating a Student class, using constructors, the this keyword, and constructor overloading

How to do it -

1. Create a class called Students that will have both variables and constructors.
2. In the class I will declare some variables.
3. And copy them to the constructor using this keyword.
4. One extra variable for showing constructor overloading
5. Another function to print all at once
6. Now take all the inputs
7. After that, just create an object and print it.

```
//Topic Covered : class, constructors, this keyword, and constructor overloading

import java.util.Scanner;

class Student
{
    int id;
    String name ;
    int section;
    String course;
    int birth ;

    Student(int id,String name, int section, String course){
        this.name = name;
        this.course = course;
        this.id = id;
        this.section = section;
        this.birth = birth;
    }

    void display()
    {
        System.out.println("ID    : " + id);
        System.out.println("NAME    : " + name);
        System.out.println("COURSE : " + course);
        System.out.println("SECTION : " + section);
        System.out.println("BIRTHDAY : " + birth);
    }
}
```

```
public class Main
{
    public static void main(String[] args)
    {

        Scanner input = new Scanner(System.in);

        System.out.print("Please Enter Unique ID of The Student: ");
        int id = input.nextInt();

        input.nextLine();

        System.out.println("Enter Student's Name: ");
        String name = input.nextLine();

        System.out.println("what is the section of the student: ");
        int section = input.nextInt();

        input.nextLine();

        System.out.println("Course Name Please: ");
        String course = input.nextLine();


        Student s1 = new Student(id, name, section, course);

        s1.display();

    }

}
```

Output-

```
ID      : 324
NAME    : Bayazid Hossain
COURSE  : Java
SECTION : 6
BIRTHDAY : 0
```

Problem 02

- **Methods in Java, including instance methods and static methods**

How to do it -

To solve this problem 1st we need proper knowledge about methods, there are 2 types of methods.

1. **instance and**
2. **Static method**

so, this are the difference between both methods

Instance Method	Static Method
Belongs to an object	Belongs to the class
Requires an object	No object required
Can use instance variables directly	Cannot directly use instance variables
Called using <code>object.method()</code>	Called using <code>ClassName.method()</code>

How to do it -

Using this idea I have to follow these steps,

1. Create A Student Class.
2. Take some variables and give variable as well.
3. Create both **Instance** and **Static Methods**.
4. Now in the main function I will just call the Static Method since it doesn't need any object to create first.
5. Then before calling the instance method I'll have to create an object first since it requires.
6. Now I'll call that instance method (Function).

```
//Topic Covered : Instance Method , Static Method

class Students{

    int id = 324;
    String name = "Bayazid" ;

    void display(){
        System.out.println("YOUR UNIQUE ID  :" +id);
        System.out.println("YOUR NAME  :" +name);
    }

    static void greetings(){
        System.out.println(" WELCOME TO THE UMS ");
    }
}

public class Methods {
    public static void main(String[] args){

        System.out.println();
        Students.greetings(); //Calling Static Methods

        System.out.println("*****");

        Students s1 = new Students();
        s1.display(); //Calling Instance Methods
    }
}
```

Output-

```
WELCOME TO THE UMS
*****
YOUR UNIQUE ID  :324
YOUR NAME       :Bayazid
```

Problem 03

- **Passing objects as an array and performing basic operations on student records**

How to do it -

1. Create a Student class with the required variables, a constructor, and a method to display the student information.
2. Create an array of Student objects with the required size.
3. Create Student objects and store each object in the array.
4. Use a loop (such as an enhanced for loop) to access each object in the array.
5. Perform the required operation on each student record by calling the appropriate method.

```
//Topic Covered : Passing objects as an array

//import java.util.Scanner;
class student
{
    int id;
    String name ;
    double cgpa;
    //int section;
    //String course;

    student(int id,String name,double cgpa){
        this.id = id;
        this.name = name;
        this.cgpa = cgpa;
    }

    void display(){

        System.out.println(id + " - " + name + " - CaGPA: " + cgpa);
    }
}

public class ObjectsNarray {
    static void main(String[] args){

        /*  Scanner input = new Scanner(System.in);

        double gpa;
        double cgpa;
        double sum=0;
        for (int i=0;i<=2;i++){
            System.out.println("Enter Three GPA for one Student :");
            gpa = input.nextInt();
            sum = sum+gpa;
        }

        cgpa = sum/3 ;*/

        student[] details = new student[3];
        details[0] = new student(324,"Bayazid",3.85);
        details[1] = new student(331,"Samia ",3.78);
        details[2] = new student(346,"Mahidur",3.45);

        System.out.println("Students Details");

        for (student s : details) {

            s.display();

        }
    }
}
```

Output-

Students Details

324 - Bayazid - CGPA: 3.85

331 - Samia - CGPA: 3.78

346 - Mahidur - CGPA: 3.45

Resources

- Idea Taken from ChatGPT/Perplexity
- PDF provided in the class room.
- Codes are completely own made

Prepared by Bayazid Hossain Parvez